Product Requirements Document (PRD)

Project Title: Digital Twin for University Building for Occupancy-aware Energy and Indoor Resource Management

1. Product Overview

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  Product Name: CampusTwin EnergyMind
- **One-Line Definition:** An occupancy-aware Digital Twin web application providing real-time room visibility, energy tracking, and resource waste detection for university buildings.
- **Product Vision:** To transform university facilities into sustainable, intelligent environments by aligning energy consumption and resource allocation dynamically with actual human occupancy.
- **How the Digital Twin Works:** The system ingests room-level occupancy and energy/resource metrics (via IoT sensors, APIs, or simulated datasets), updates a digital 2D spatial representation of the building, correlates human presence with active loads (HVAC, lighting), and highlights energy wastage on a central administrator dashboard.
- **Key Value Proposition:** Eliminates hidden energy waste in empty university spaces, improves classroom/lab utilization transparency, and equips facility managers with real-time actionable insights without requiring manual physical inspections.

2. Problem Statement

Universities manage expansive physical infrastructures with highly dynamic daily schedules. However, traditional facility operations suffer from several critical inefficiencies:

- **Lack of Real-Time Visibility:** Administrators lack instant visibility into which classrooms, lecture halls, or laboratories are currently occupied or vacant.
- **Energy Consumption in Unoccupied Spaces:** HVAC systems, lights, projectors, and workstation power frequently remain fully active in empty rooms due to static timer schedules or human oversight.
- **Inefficient Space Utilization:** Rooms are often reserved on paper but left empty in practice, leading to perceived space shortages across campus.
- **Manual Monitoring Overhead:** Operations staff must physically audit floors and rooms to verify resource status and room availability.
- **Fragmented Building Data:** Energy metrics, room schedules, and resource states exist in

isolated silos, making cross-correlation and waste identification difficult.

Why an Occupancy-Aware Digital Twin Helps: By visually mapping telemetry data directly onto a spatial digital model of the building, the Digital Twin provides real-time situational awareness. Correlating occupancy with active power loads allows immediate identification of off-schedule or wasteful energy consumption.

3. Target Users

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User Role	Primary Needs & Responsibilities	System Interaction
University Facility / Building Administrators	Needs high-level visibility over building operational efficiency, energy costs, and room utilization trends across departments.	Accesses executive dashboards, reviews aggregated utilization reports, monitors major energy anomalies.
Energy / Resource Managers	Responsible for energy conservation targets, sustainability compliance, and detecting specific hardware or operational waste.	Monitors real-time energy vs. occupancy correlation views, reviews automated waste alerts, analyzes trend graphs.
Campus Operations Staff	Handles day-to-day physical maintenance, room inspections, and turning off left-on equipment.	Receives operational waste alerts (e.g., "Room 302 empty, HVAC ON"), targets physical inspections directly to flag areas.
Faculty & Students (Secondary Users)	Needs transparent visibility into real-time study space and available room capacity.	Accesses a read-only view of room availability and current occupancy levels.

4. Product Goals

Measurable MVP Goals

- Provide a digital 2D floor-by-floor layout representation of a selected university building.
- Monitor and visually reflect real-time occupancy status (Occupied vs. Unoccupied) across 100% of mapped rooms.
- Display energy consumption (kWh / active wattage) and indoor resource status (Lighting: ON/OFF, HVAC: Active/Inactive).
- Identify and flag 100% of simulated or detected instances where energy is consumed above threshold in empty rooms.
- Deliver a central dashboard with telemetry refresh latency of 5 seconds or less for simulated/ingested streams.
- Provide historical data visualization (24-hour, 7-day, 30-day) comparing occupancy patterns against energy usage.

Non-Goals (Explicitly Excluded from MVP)

- Direct, autonomous control of physical HVAC, breakers, or lighting hardware (monitoring only).
- Individual person tracking, facial recognition, or personal identity capture.
- Complex 3D graphics, WebGL rendering, or Virtual Reality (VR) environments.
- Multi-building or campus-wide federation (focus is on a single representative building).

5. Core Features

5.1 Digital Twin Building Visualization

- Interactive 2D spatial layout of building floors, rooms, classrooms, and labs.
- Dynamic color-coded status overlays per room (e.g., Green = Occupied/Normal, Red = Waste Alert, Gray = Unoccupied/Idle).
- Clickable room nodes displaying granular metadata: room ID, capacity, current occupancy, lighting state, HVAC state, and current energy draw.

5.2 Occupancy Monitoring

- Display current room status: *Occupied*, *Unoccupied*, *Available*.
- Headcount display for spaces equipped with count-capable data streams.
- Historical hourly occupancy breakdown.

5.3 Energy Monitoring

- Room-level and floor-level energy draw (kW/kWh).
- Visual indicator comparing actual energy load against baseline standby load.
- Energy trend overlays for peak vs. off-peak operational hours.

5.4 Occupancy-Aware Energy Insights

- Automated correlation engine detecting active loads in zero-occupancy spaces.
- Energy waste alert generation (e.g., "HVAC running in empty Room 204 for > 30 minutes").
- Quantified potential energy cost/kWh waste metrics per event.

5.5 Indoor Resource Management

- Status indicators for HVAC/AC units (Active, Idle, OFF) and Lighting banks (ON, OFF, Dimmed).
- Real-time room availability status based on combined calendar schedule and actual physical occupancy.

5.6 Administrator Dashboard

- Summary metrics cards: Total Building Load, Current Building Occupancy %, Active Waste Alerts, Energy Wasted Today (kWh).
- Filterable floor view and active alerts panel.
- Exportable summaries for resource reporting.

5.7 Historical Data & Analytics

- Occupancy vs. Energy overlay charts over custom date ranges.
- Space utilization heatmaps detailing most and least utilized rooms.

6. MVP Scope

In Scope (MVP Deliverables)	Out of Scope (Future Enhancements)
Digital 2D layout map for 1 university building (3-5 floors).	Full campus-wide 3D / VR building renders.
Real-time room occupancy & energy status visualization.	Automated/autonomous control of physical hardware switches.

In Scope (MVP Deliverables)	Out of Scope (Future Enhancements)
Simulated data generator / REST API ingestion for IoT streams.	Biometric or facial recognition tracking.
Rule-based energy waste detection engine & alerts.	Advanced AI predictive maintenance & machine learning forecasting.
Admin dashboard with historical trend analytics & room utilization charts.	Native mobile apps (iOS/Android) — Web application only.

7. User Stories

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  US-01 (Facility Admin): As a Facility Administrator, I want to view a digital floor map showing live room occupancy, so that I can immediately understand space utilization across campus.
- **US-02 (Energy Manager):** As an Energy Manager, I want to see automated alerts when HVAC or lighting is running in empty rooms, so that I can eliminate unnecessary power waste.
- **US-03 (Operations Staff):** As an Operations Staff member, I want a list of active energy-waste locations prioritized by duration, so that I can efficiently target physical checks.
- **US-04 (Facility Admin):** As a Facility Administrator, I want to analyze 30-day historical occupancy against energy consumption, so that I can adjust building operational schedules based on real data.
- **US-05 (Faculty/Student):** As a Student, I want to check room availability status remotely, so that I can find an empty study room without wandering through hallways.

8. Functional Requirements

Req ID	Feature Category	Functional Requirement Description
FR-01	Spatial Representation	The system shall render an interactive 2D graphical representation of building floors and individual room boundaries.
FR-02	Occupancy Display	The system shall display live occupancy status (Occupied / Unoccupied) and current headcounts (where available) per room.
FR-03	Energy Data Ingestion	The system shall ingest or simulate real-time energy usage data (kW/kWh) at both room and floor levels.
FR-04	Resource Monitoring	The system shall track and display operational states of Lighting (ON/OFF) and HVAC (Active/Idle/OFF) per room.
FR-05	Waste Correlation	The system shall trigger an energy-waste condition when a room's occupancy is zero while HVAC or Lighting load exceeds predefined standby thresholds for over 15 minutes.
FR-06	Alert Generation	The system shall generate visual dashboard alerts

Req ID	Feature Category	Functional Requirement Description
		containing room location, waste type, duration, and estimated wasted kWh.
FR-07	Dashboard Aggregation	The system shall display total building energy rate, total occupancy percentage, and count of active alerts on an executive overview.
FR-08	Historical Analytics	The system shall provide time-series line and bar charts comparing hourly/daily occupancy trends against energy consumption.
FR-09	Data Simulation Engine	The system shall provide a controllable data simulation module to stream synthetic occupancy and telemetry data for demo and testing purposes.
FR-10	Role-Based Access	The system shall restrict administrative dashboard features and rule configuration to authenticated administrative users.

9. Non-Functional Requirements

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Performance: Dashboard UI updates within 2 seconds of state change ingestion. Historical chart rendering time shall not exceed 1.5 seconds for 30-day datasets.

- **Scalability:** System architecture must support ingesting telemetry from up to 500 rooms simultaneously at 5-second sampling intervals.
- **Reliability & Availability:** The web application shall maintain 99.0% uptime during operational building hours (07:00 to 22:00).
- **Security:** All communication protocols must enforce TLS 1.3 encryption. Role-based access control (RBAC) must protect admin endpoints.
- **Privacy:** System design strictly avoids collecting Personally Identifiable Information (PII). Occupancy data is aggregated strictly as numerical headcounts or binary presence.
- **Usability:** UI must follow clean responsive design principles usable on standard desktop resolutions (1920x1080 and 1366x768).

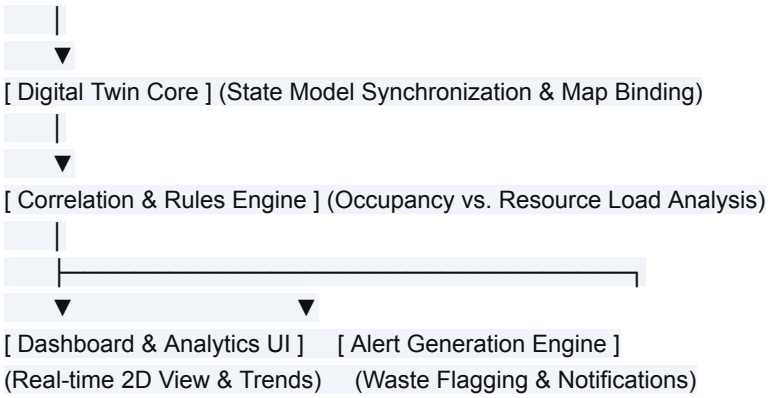
10. Data Requirements

Core Data Entities

- **Building:** ID, Name, Total Floors, Address, Gross Area (sq ft).
- **Floor:** ID, Building_ID, Floor Number, Floor Map SVG/JSON Layout Data.
- **Room:** ID, Floor_ID, Room Number, Room Type (Classroom, Lab, Office), Max Capacity, Standby Power Threshold (kW).
- **Occupancy Stream:** Stream_ID, Room_ID, Timestamp, IsOccupied (Boolean), OccupantCount (Integer).
- **Energy Stream:** Stream_ID, Room_ID, Timestamp, ActivePower (kW), EnergyConsumed (kWh).
- **Resource State:** State_ID, Room_ID, Timestamp, LightingState (ON/OFF), HVACState (Active/Idle/OFF).
- **Alert Record:** Alert_ID, Room_ID, Timestamp, AlertType, Severity, Status (Active/Resolved), EstimatedWastedkWh.

11. System Workflow





12. User Interface Requirements

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  Executive Header: Live status counters showing Building Energy Rate (kW), Occupancy Rate (%), Active Waste Alerts, and System Health.
- **Interactive Building Explorer:** Left panel for floor navigation; center stage showing interactive 2D floor plans with room color coding.
- **Room Detail Modal/Drawer:** Slide-out drawer on room click showing real-time gauge of power draw, HVAC status, headcount, and 24-hour graph.
- **Alert Center Table:** Filterable table showing Room #, Alert Trigger Time, Issue Type, Wasted kWh, and "Acknowledge" action button.
- **Analytics Tab:** Dual-axis line charts superimposing Occupancy (line) over Power Consumption (bar) across selectable date ranges.

13. Success Metrics

KPI Category	Metric Description	Proposed MVP Target
Data Representation	Percentage of target building rooms correctly mapped in Digital Twin	100% (Target: 1 Building, ~30-50 spaces)
Waste Detection	Detection accuracy rate of simulated energy waste	≥ 95%

KPI Category	Metric Description	Proposed MVP Target
	scenarios	
System Speed	Dashboard render & updates response latency	< 2 seconds
Usability	Facility staff task completion rate for identifying waste rooms	100% within 3 clicks
System Availability	Platform uptime during lab testing and validation phase	≥ 99%

14. Assumptions

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  Occupancy and energy data for the MVP can be supplied via a synthetic data generator mimicking realistic university schedules if real sensor hardware is unavailable.
- The MVP implementation is constrained to 1 representative university building (up to 4 floors).
- Physical direct control of HVAC/Lighting is out of scope; recommendations are operational for human staff intervention.
- Administrative users access the platform via standard modern web browsers (Chrome, Edge, Firefox).

15. Risks and Mitigation

Identified Risk	Impact	Practical Mitigation Strategy
Missing or noisy IoT hardware data stream.	High	Implement a robust mock telemetry generator with realistic daily curves to

Identified Risk	Impact	Practical Mitigation Strategy
		guarantee full software testing.
User privacy concerns regarding room tracking.	High	Strictly enforce anonymous count-only data structures; avoid camera/facial data integration entirely.
False-positive energy waste alerts.	Medium	Implement configurable duration hysteresis (e.g., room must be empty for 15+ continuous minutes before alerting).
Complex spatial file integration delays.	Medium	Use simplified clean SVG/JSON floorplan layouts rather than heavy CAD/BIM files for MVP frontend rendering.

16. Acceptance Criteria

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  AC-01 (Digital Twin Mapping): Given an authenticated user on the dashboard, when they select "Floor 2", then the system displays the interactive 2D layout of Floor 2 with all rooms rendered in their current status colors within 1 second.
- **AC-02 (Occupancy Update):** Given a change in room occupancy state, when the data ingestion endpoint receives the update, then the room color and numerical count on the dashboard update within 2 seconds.
- **AC-03 (Waste Detection Alert):** Given Room 101 has 0 occupancy and HVAC power load > 1.5 kW for over 15 minutes, when the correlation engine runs, then a high-priority "Energy Waste Alert" appears in the alert panel highlighting Room 101.
- **AC-04 (Historical Analytics Filtering):** Given an energy manager on the Analytics view, when they select a "Last 7 Days" filter for Room 301, then a dual-axis chart renders

showing hourly occupancy vs. energy consumption.

17. MVP Output Scope

The final MVP output will deliver a fully functional software prototype comprising:

- Web-based front-end dashboard with interactive 2D Digital Twin building and floor visualizations.
- Backend API service handling telemetry ingestion, state management, and rule correlation.
- Built-in configurable university building dataset simulator (generating realistic lecture/lab usage cycles).
- Alerting and historical analytics engine with exportable room utilization summaries.

18. Future Enhancements

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  Integration with actual physical Building Management Systems (BMS) via BACnet/Modbus for direct HVAC/lighting control.
- AI-driven occupancy forecasting to pre-cool or pre-heat spaces based on timetable schedules.
- Integration with university timetabling software (e.g., Canvas, Banner) to compare booked vs. actual usage.
- Campus-wide multi-building Digital Twin federation.
- Carbon emission footprint tracking (CO_2 output per space per day).

19. Out of Scope

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  Biometrics, facial recognition, or individual student/staff tracking.
- Autonomous physical system control (switching off physical breakers or valves automatically).
- Native iOS/Android mobile application development.
- Virtual Reality (VR) / Augmented Reality (AR) spatial walkthroughs.

20. Product Constraints and Dependencies

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  Dependencies: Availability of digital CAD/SVG floor layout assets for mapping; access to network infrastructure for REST/WebSocket communication.

- **Constraints:** Web-only deployment target; simulation-reliant if physical campus IoT endpoints are restricted by IT policy.

MVP Summary

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  Problem Being Solved: Energy wastage in unoccupied university spaces and lack of spatial utilization visibility for campus facilities.
- **Target Users:** University Facility Administrators, Energy Managers, and Campus Operations Staff.
- **MVP Solution:** A web-based 2D Digital Twin application providing real-time room occupancy and energy correlation with automated waste detection.
- **Core Features:** 2D Digital Twin Visualization, Live Occupancy & Energy Tracking, Waste Alert Engine, Admin Dashboard, Historical Analytics.
- **Key Success Metrics:** 100% target space mapping, ≥ 95% waste detection accuracy, < 2-second UI refresh response time.
- **Major Exclusions:** Autonomous hardware control, facial recognition, multi-building campus scope, 3D/VR rendering.
- **Expected Deliverable:** Fully functional software prototype with dashboard UI, analytics engine, and configurable telemetry simulator.